COMPLEX NUMBER REVIEW

Given that $z = e^{j\theta}$ and $z \neq 1$,

Show that

$$\left|\frac{z^n - 1}{z - 1}\right| = \frac{\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}, \text{ and find } \arg\left(\frac{z^n - 1}{z - 1}\right)$$

▼ Solution

$\frac{z^n-1}{z-1} = \frac{e^{jn\theta}-1}{e^{j\theta}-1}$ (directly use definition of z)

$= \frac{e^{\frac{2jn\theta}{2}}-1}{e^{\frac{2j\theta}{2}}-1}$ (since they want theta/2, better to keep everything in terms of theta/2)

$= \frac{e^{\frac{jn\theta}{2}} \times e^{\frac{jn\theta}{2}}-1}{e^{\frac{j\theta}{2}} \times e^{\frac{j\theta}{2}}-1}$ (see if can isolate ntheta/2 and theta/2 to get closer to answer)

$= \frac{e^{\frac{jn\theta}{2}} \times e^{\frac{jn\theta}{2}} - e^{\frac{jn\theta}{2}} \times e^{-\frac{jn\theta}{2}}}{e^{\frac{j\theta}{2}} \times e^{\frac{j\theta}{2}} - e^{\frac{j\theta}{2}} \times e^{-\frac{j\theta}{2}}}$ (again, since answer is only in terms of theta/2, best to convert)

$= \frac{e^{\frac{jn\theta}{2}}\left(e^{\frac{jn\theta}{2}} - e^{-\frac{jn\theta}{2}}\right)}{e^{\frac{j\theta}{2}}\left(e^{\frac{j\theta}{2}} - e^{-\frac{j\theta}{2}}\right)}$ (see common terms)

$= \frac{e^{\frac{jn\theta}{2}}}{e^{\frac{j\theta}{2}}} \times \frac{e^{\frac{jn\theta}{2}} - e^{-\frac{jn\theta}{2}}}{e^{\frac{j\theta}{2}} - e^{-\frac{j\theta}{2}}}$

$= \frac{e^{\frac{jn\theta}{2}}}{e^{\frac{j\theta}{2}}} \times \frac{\cos\left(\frac{n\theta}{2}\right) + j\sin\left(\frac{n\theta}{2}\right) - \left(\cos\left(\frac{n\theta}{2}\right) - j\sin\left(\frac{n\theta}{2}\right)\right)}{\cos\left(\frac{\theta}{2}\right) + j\sin\left(\frac{\theta}{2}\right) - \left(\cos\left(\frac{\theta}{2}\right) - j\sin\left(\frac{\theta}{2}\right)\right)}$ (see theta/2 and -theta/2 → likely to cancel)

$= \frac{e^{\frac{jn\theta}{2}}}{e^{\frac{j\theta}{2}}} \times \frac{\cos\left(\frac{n\theta}{2}\right) + j\sin\left(\frac{n\theta}{2}\right) - \cos\left(\frac{n\theta}{2}\right) + j\sin\left(\frac{n\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right) + j\sin\left(\frac{\theta}{2}\right) - \cos\left(\frac{\theta}{2}\right) + j\sin\left(\frac{\theta}{2}\right)}$

$= \frac{e^{\frac{jn\theta}{2}}}{e^{\frac{j\theta}{2}}} \times \frac{2j\sin\left(\frac{n\theta}{2}\right)}{2j\sin\left(\frac{\theta}{2}\right)}$

$= e^{j\left(\frac{n\theta}{2} - \frac{\theta}{2}\right)} \times \frac{\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$ (answer now obtained, now show this term will be the magnitude)

$= \frac{\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)} \angle \frac{\theta}{2}(n-1)$

Thus, $\left|\frac{z^n-1}{z-1}\right| = \left|\frac{\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}\angle\frac{\theta}{2}(n-1)\right| = \frac{\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$

and $\arg\left(\frac{z^n-1}{z-1}\right) = \frac{\theta}{2}(n-1)$.

Given that $w\bar{w} = 4$ and $\arg\left(\frac{2\bar{w}}{j}\right) = -\frac{2\pi}{3}$

Find w, and hence find the smallest value of w such that $w^n \in \mathbb{R}$

▼ Solution

Let $w = r\angle\theta$

$w\bar{w} = 4$ (standard that $z\bar{z} = a^2 + b^2 = r^2$)

$r^2 = 4$

$r = 2$

$\frac{2\bar{w}}{j} = \frac{2r\angle-\theta}{1\angle\frac{\pi}{2}}$ (isolate angle to easily use arg() function)

$= 2r\angle-\theta-\frac{\pi}{2}$

$\therefore \arg\left(\frac{2\bar{w}}{j}\right) = -\frac{2\pi}{3}$

$-\theta - \frac{\pi}{2} = -\frac{2\pi}{3}$

$\frac{2\pi}{3} - \frac{\pi}{2} = \theta$

$\theta = \frac{\pi}{6}$

$\therefore w = 2\angle\frac{\pi}{6}$

Thus,

$w^n = 2^n\angle\frac{n\pi}{6}$

For $w^n \in \mathbb{R}$,

$\frac{n\pi}{6} = \pi$

$n = 6$

Given that $w = a + jb, a \in \mathbb{R}, b \in \mathbb{R}, > 10$, find w given

$$\left|\frac{w-12}{w-j8}\right| = \frac{5}{3}, \ \left|\frac{w-4}{w-8}\right| = 1$$

▼ Solution

$\left|\frac{w-4}{w-8}\right| = 1$ (brute force, see what works, pick =1 as most likely easier equation)

$\left|\frac{a+jb-4}{a+jb-8}\right| = 1$

$\sqrt{\frac{(a-4)^2+b^2}{(a-8)^2+b^2}} = 1$

$\frac{(a-4)^2+b^2}{(a-8)^2+b^2} = 1^2$

$\frac{a^2+b^2-8a+16}{a^2+b^2-16a+64} = 1$ (notice same a^2+b^2)

$\cancel{a^2+b^2} - 8a + 16 = \cancel{a^2+b^2} - 16a + 64$

$-8a + 16 = -16a + 64$ (equation is in terms of a only)

$16a - 8a + 16 = 64 - 16$

$8a = 48$

$a = 6$ (similar concept can be used to find b by substituting in a first)

$\left|\frac{w-12}{w-j8}\right| = \frac{5}{3}$

$\left|\frac{a+jb-12}{a+jb-j8}\right| = \frac{5}{3}$

$\sqrt{\frac{(a-12)^2+b^2}{a^2+(b-8)^2}} = \frac{5}{3}$

$\frac{(a-12)^2+b^2}{a^2+(b-8)^2} = \left(\frac{5}{3}\right)^2$

$\frac{(6-12)^2+b^2}{6^2+(b-8)^2} = \left(\frac{5}{3}\right)^2$

$\frac{36+b^2}{36+b^2-16b+64} = \frac{25}{9}$

$\frac{36+b^2}{b^2-16b+100} = \frac{25}{9}$

$9(36 + b^2) = 25(b^2 - 16b + 100)$

$324 + 9b^2 = 25b^2 - 400b + 2500$

$9b^2 - 25b^2 + 400b - 2500 + 324 = 0$

$-16b^2 + 400b - 2176 = 0$

$-16(b - 8)(b - 17) = 0$

$b = 17(\text{rej. } 8 \text{ as } b > 10)$

$\therefore w = 6 + j17$

Given that $z \in \mathbb{C}$, where $\arg(z) = \frac{-5\pi}{9}$,

Show that

$$\frac{z - \bar{z}}{1 + z\bar{z}} \in \mathbb{I}, \text{ and find } \arg\left(\frac{z - \bar{z}}{1 + z\bar{z}}\right)$$

▼ Solution

Let $z = r\angle - \frac{5\pi}{9}$

$\therefore \bar{z} = r\angle\frac{5\pi}{9}$

Thus,

$\frac{z-\bar{z}}{1+z\bar{z}} = \frac{r\angle-\frac{5\pi}{9} - r\angle\frac{5\pi}{9}}{1+r^2}$ (use given information to sub in, notice that denom is purely real)

$= \frac{r\cos(\frac{5\pi}{9}) - jr\sin(\frac{5\pi}{9}) - (r\cos(\frac{5\pi}{9}) + jr\sin(\frac{5\pi}{9}))}{1+r^2}$ (thus numerator will be purely imaginary)

$= \frac{\cancel{r\cos(\frac{5\pi}{9})} - jr\sin(\frac{5\pi}{9}) - \cancel{r\cos(\frac{5\pi}{9})} - jr\sin(\frac{5\pi}{9})}{1+r^2}$

$= \frac{-j2r\sin(\frac{5\pi}{9})}{1+r^2}$

$= -j\left(\frac{2r\sin(\frac{5\pi}{9})}{1+r^2}\right)$

Thus, $\frac{z-\bar{z}}{1+z\bar{z}}$ is purely imaginary.

As $\arg(-j) = -\frac{\pi}{2}$,

$\arg\left(-j\left(\frac{2r\sin(\frac{5\pi}{9})}{1+r^2}\right)\right) = -\frac{\pi}{2}$

Show that

$$(a + jb)^{4n} + (b + ja)^{4n} \in \mathbb{R}, \text{ given } (a, b) \in \mathbb{R}, n \in \mathbb{N}$$

▼ Solution

$(a + jb)^{4n} + (b + ja)^{4n} = (a + jb)^{4n} + j^{4n}(b + ja)^{4n}$ (1 = 1^n= (j^4)^n = j^4n)

$= (a + jb)^{4n} + j^{4n}(b + ja)^{4n}$ (multiplying by j to b+ja makes a real, b imaginary)

$= (a + jb)^{4n} + (jb + j^2 a)^{4n}$

$= (a + jb)^{4n} + (-a + jb)^{4n}$ (notice that we can factorise out -1 to get +a on both terms)

$= (a + jb)^{4n} + (-(a - jb))^{4n}$

$= (a + jb)^{4n} + (a - jb)^{4n}$ (likely to cancel)

Let $a + jb = r\angle\theta$. Thus, $a - jb = r\angle -\theta$.

$(a + jb)^{4n} + (a - jb)^{4n} = (r\angle\theta)^{4n} + (r\angle -\theta)^{4n}$

$= r\angle 4n\theta + r\angle -4n\theta$

$= r\cos(4n\theta) \cancel{+ jr\sin(4n\theta)} + r\cos(4n\theta) \cancel{- jr\sin(4n\theta)}$

$= 2r\cos(4n\theta)$

Thus $(a + jb)^{4n} + (b + ja)^{4n}$ is purely real.

Given that $z \in \mathbb{C}$, find z given that

$$|z - 3| = 4,\ \arg(z) = \frac{3\pi}{4}$$

▼ Solution

As $\arg(z) = \frac{3\pi}{4}$, let $z = -a + ja$ where $a > 0$. Thus,

$|z - 3| = 4$

$|-a + ja - 3| = 4$

$(-3 - a)^2 + a^2 = 4^2$

$9 + 2(-3)(-a) + a^2 + a^2 = 16$

$2a^2 + 6a - 7 = 0$

$a = \frac{-6 \pm \sqrt{6^2 - 4(2)(-7)}}{2(2)}$

$= -3.90, 0.898(\text{rej. -3.90 as } a > 0)$

$\therefore z = -0.898 + j0.898$

Express Z in polar form, where

$$Z = (\sin(\theta) + j\cos(\theta))^3$$

▼ Solution

$Z = j^4(\sin(\theta) + j\cos(\theta))^3$

$= jj^3(\sin(\theta) + j\cos(\theta))^3$ (again, we want to make cosine the real term, so we multiply by j)

$= j(j\sin(\theta) + j^2\cos(\theta))^3$

$= j(j\sin(\theta) - \cos(\theta))^3$

$= j(-\cos(\theta) + j\sin(\theta))^3$

$= j(-(\cos(\theta) - j\sin(\theta)))^3$

$= -j(\cos(\theta) - j\sin(\theta))^3$

$= (1\angle -\frac{\pi}{2})(1\angle -\theta)^3$

$= (1\angle -\frac{\pi}{2})(1\angle -3\theta)$

$= 1\angle(-\frac{\pi}{2} - 3\theta)$

Given that $z = r(\cos(\theta) + j\sin(\theta))$, where $r > 0$, show that

$$\frac{r + \cos(\theta) + j\sin(\theta)}{1 + r\cos(\theta) - jr\sin(\theta)} = \frac{z}{r}$$

▼ Solution

$\frac{r+\cos(\theta)+j\sin(\theta)}{1+r\cos(\theta)-jr\sin(\theta)} = \frac{r+\frac{z}{r}}{1+\bar{z}}$ ($r\angle -\theta = r\bar{\angle}\theta$) (express everything using the given definitions)

$= \frac{r+\frac{z}{r}}{1+\frac{r^2}{z}}$ (we want to get rid of the fraction in fraction stuff)

$= \frac{\frac{r^2}{r}+\frac{z}{r}}{\frac{z}{z}+\frac{r^2}{z}}$

$= \frac{\frac{r^2+z}{r}}{\frac{z+r^2}{z}}$ (we notice same r^2+z)

$= \frac{\frac{1}{r}}{\frac{1}{z}}$

$= \frac{z}{r}$

Given that $z, w \in \mathbb{C}$, find z and w in rectangular form given

$$z\bar{z} = 2,\ \arg(-z) = \arg(w) = -\frac{\pi}{4},\ |z + w| = 5\sqrt{2}$$

▼ Solution

$z\bar{z} = 2$

$|z| = \sqrt{2}$ (standard way to find magnitude

Let $z = \sqrt{2}\angle\theta$

$\therefore -1 \times z = 1\angle\pi \times \sqrt{2}\angle\theta$ (arg -z is given, use to directly find z first)

$= \sqrt{2}\angle\pi + \theta$

As $\arg(-z) = -\frac{\pi}{4}$

$\pi + \theta = -\frac{\pi}{4}$

$-\theta = \pi + \frac{\pi}{4}$

$\theta = -\frac{5\pi}{4}$

$\text{Thus, } z = \sqrt{2}\angle 2\pi - \frac{5\pi}{4} = \sqrt{2}\angle\frac{3\pi}{4} = -1 + j$ (prep as rectangular form for z+w later)

$\text{As } \arg(w) = -\frac{\pi}{4}$, (draw argand diagram, notice how w is a 1 to 1 diagonal)

$\text{Let } w = a - ja$, where $a > 0$ (have to add a>0 otherwise arg(w) will not apply)

$\therefore |-1 + j + a - ja| = 5\sqrt{2}$ (can add and sovle)

$\sqrt{(-1+a)^2 + (1-a)^2} = 5\sqrt{2}$

$(-1+a)^2 + (1-a)^2 = (5\sqrt{2})^2$ (notice similar a-1 and 1-a)

$(-1)^2(-1+a)^2 + (1-a)^2 = (5\sqrt{2})^2$

$(1-a)^2 + (1-a)^2 = (5\sqrt{2})^2$

$2(1-a)^2 = 50$

$(1-a)^2 = 25$

$1 - a = \pm 5$

$-a = \pm 5 - 1$

$a = 1 \pm 5$

$= 6, -4(\text{rej -4 as } a > 0)$

$\therefore z = -1 + j, w = 6 - 6j$

Simplify $(z^n - e^{j\theta})(z^n - e^{-j\theta})$. Hence, solve

$$z^6 - \sqrt{3}z^3 + 1 = 0$$

and express z in exponential form, where $z \in \mathbb{C}$ and lies in the fourth quadrant.

▼ Solution

$(z^n - e^{j\theta})(z^n - e^{-j\theta}) = z^{2n} - z^n e^{-j\theta} - z^n e^{j\theta} + 1$ (expand, -jtheta&jtheta powers cancel)

$= z^{2n} - z^n \left(e^{-j\theta} + e^{j\theta}\right) + 1$ (factorise)

$z^6 - \sqrt{3}z^3 + 1 = 0$

$(z^3)^2 - \sqrt{3}z^3 + 1 = 0$ (notice similarity)

$\text{By comparison with } z^{2n} - z^n \left(e^{-j\theta} + e^{j\theta}\right) + 1 \text{ (when } n = 3\text{)},$

$\left(e^{-j\theta} + e^{j\theta}\right) = \sqrt{3}$ (feels like it will cancel in trig form/ conjugates added together)

$\cos(\theta) \cancel{- j\sin(\theta)} + \cos(\theta) \cancel{+ j\sin(\theta)} = \sqrt{3}$

$2\cos(\theta) = \sqrt{3}$

$\cos(\theta) = \frac{\sqrt{3}}{2}$

$\theta = \frac{\pi}{6}$ (Full range of theta is not expected in ESTs)

thus,

$(z^3)^2 - \sqrt{3}z^3 + 1 = 0$ (notice how original form is factorised)

$(z^3 - e^{j\frac{\pi}{6}})(z^3 - e^{-j\frac{\pi}{6}}) = 0$

$z^3 = e^{j\frac{\pi}{6}}, e^{-j\frac{\pi}{6}}$

$z = e^{j\frac{\pi}{18}}, e^{-j\frac{\pi}{18}}$

$z = e^{-j\frac{\pi}{18}}$ (rej. positive power as z lies in fourth quadrant.)

Given that $z = \cos(\theta) + j\sin(\theta)$,

express $\cos(k\theta)$ in terms of z^k and $\frac{1}{z^k}$, and express $\sin(k\theta)$ in terms of z^k and $\frac{1}{z^k}$.

Hence, show that

$$\cos^2(\theta)\sin^2(\theta) = -\frac{1}{16}\left(z^2 - \frac{1}{z^2}\right)^2$$

▼ Solution

$z^k = \cos(k\theta) + j\sin(k\theta) = 1\angle k\theta$ (breakdown z^k)

$\frac{1}{z^k} = \frac{1}{1\angle k\theta} = 1\angle -k\theta$ (breakdown z^-k)

$= \cos(k\theta) - j\sin(k\theta)$

$z^k + \frac{1}{z^k} = \cos(k\theta) + j\sin(k\theta) + \cos(k\theta) - j\sin(k\theta)$ (notice + and - j terms)

$z^k + \frac{1}{z^k} = 2\cos(k\theta)$

$\cos(k\theta) = \frac{1}{2}\left(z^k + \frac{1}{z^k}\right)$

$z^k - \frac{1}{z^k} = \cos(k\theta) + j\sin(k\theta) - \cos(k\theta) + j\sin(k\theta)$ (try opposite, minus)

$z^k - \frac{1}{z^k} = 2j\sin(k\theta)$

$\sin(k\theta) = \frac{1}{2j}\left(z^k - \frac{1}{z^k}\right)$

Let k=1, (notice how cos(theta) has k=1)

$\cos(\theta) = \frac{1}{2}\left(z + \frac{1}{z}\right)$

$\sin(\theta) = \frac{1}{2j}\left(z - \frac{1}{z}\right)$ (notice how cos(theta)sin(theta) has (a+b)(a-b) property)

Thus, $\cos^2(\theta)\sin^2(\theta) = (\cos(\theta)\sin(\theta))^2$ (reach familliarity, combine cos and sin first)

$= \left(\frac{1}{2}\left(z + \frac{1}{z}\right) \times \frac{1}{2j}\left(z - \frac{1}{z}\right)\right)^2$

$= \left(\frac{1}{4j}\left(z + \frac{1}{z}\right)\left(z - \frac{1}{z}\right)\right)^2$

$= \left(\frac{1}{4j}\right)^2 \left(\left(z + \frac{1}{z}\right)\left(z - \frac{1}{z}\right)\right)^2$

$= -\frac{1}{16}\left(z^2 - \frac{1}{z^2}\right)^2$

Given $z = -1 + j\sqrt{3}$, find the integer a such that

$$\arg(z(z+a)) = \frac{5\pi}{6}$$

▼ Solution

$z(z+a) = (-1 + j\sqrt{3})(a - 1 + j\sqrt{3})$

$= -(a-1) - j\sqrt{3} + j(a-1)\sqrt{3} - 3$

$= -a - 2 + j(a-1)\sqrt{3} - j\sqrt{3}$

$= -a - 2 + j((a-1)\sqrt{3} - \sqrt{3})$

$= -a - 2 + j\sqrt{3}((a-1) - 1)$

$= -a - 2 + j\sqrt{3}(a-2)$

$\therefore \arg(z(z+a)) = \frac{5\pi}{6}$

$\arg(-a - 2 + j\sqrt{3}(a-2)) = \frac{5\pi}{6}$

$\tan^{-1}\left(\frac{\sqrt{3}(a-2)}{-a-2}\right) = \frac{5\pi}{6}$

$\frac{\sqrt{3}(a-2)}{-a-2} = \tan\left(\frac{5\pi}{6}\right)$

$\frac{\sqrt{3}(a-2)}{-a-2} = -\frac{\sqrt{3}}{3}$

$\frac{a-2}{-a-2} = -\frac{1}{3}$

$3(a-2) = -(-a-2)$

$3a - 6 = a + 2$

$3a - a = 2 + 6$

$2a = 8$

$a = 4$

It is given that

$$z = \frac{(1+j)^3}{\sqrt{2}(a+j)^2}$$

where a>0 and |z|=0.5. Find a, arg(z) and the smallest integer n such that $z^n \in \mathbb{R}$

▼ Solution

$z = \frac{(1+j)^3}{\sqrt{2}(a+j)^2}$

$= \frac{(\sqrt{2}\angle 45^\circ)^3}{\sqrt{2}(\sqrt{a^2+1^2}\angle \tan^{-1}(\frac{1}{a}))^2}$

$= \frac{2\sqrt{2}\angle 135^\circ}{\sqrt{2}(\sqrt{a^2+1^2}\angle \tan^{-1}(\frac{1}{a}))^2}$

$= \frac{2\angle 135^\circ}{a^2+1^2\angle 2\tan^{-1}(\frac{1}{a})}$

$= \frac{2}{a^2+1^2}\angle 135^\circ - 2\tan^{-1}(\frac{1}{a})$

Thus, as $|z| = \frac{2}{a^2+1}$,

$\frac{2}{a^2+1} = 0.5$

$\frac{2}{0.5} = a^2 + 1$

$a^2 + 1 = 4$

$a = \sqrt{3}$

$\therefore \arg(z) = 135^\circ - 2\tan^{-1}(\frac{1}{a})$

$= 135^\circ - 2\tan^{-1}(\frac{1}{\sqrt{3}})$

$= 75^\circ$

$z = 0.5\angle 75^\circ$

$z^n = 0.5^n\angle 75^\circ \times n$

$Im(z^n) = 0.5^n \sin(75n^\circ)$

$0.5^n \sin(75n^\circ) = 0$

aagh should use radians smh

Let $z = r_1\angle\theta, w = r_2\angle -\frac{\pi}{4}$, where

$$z\bar{z} = 32,\ \arg(-z) = \arg(w),\ |z+w| = \sqrt{8}, < |w|$$

Find r1, r2 and theta.

▼ Solution

Given that z and w are complex numbers, where

$$\bar{z} = j\bar{w},\ \arg(zw) = -\pi$$

Find arg(z) and arg(w).

▼ Solution

Let $z = r_1\angle\theta, w = r_2\angle\phi$

$\therefore \arg(\bar{z}) = \arg(j\bar{w})$

$-\theta = \frac{\pi}{2} - \phi \rightarrow (1)$

$\arg(zw) = -\pi$

$\theta + \phi = -\pi \rightarrow (2)$

Solving 1 and 2 simulatenously,

$\arg(z) = \theta = -\frac{3\pi}{4}$

$\arg(w) = \theta = -\frac{\pi}{4}$

It is given that z is a complex number where

$$z = \frac{(\sqrt{3} - j)^3}{(1 + jx)^2}, |z| = 4, x \in \mathbb{R}^+$$

Find x and arg(z).

▼ Solution

It is given that

$$\left|\frac{z + j}{z}\right| = 4, \arg\left(\frac{z + j}{z}\right) = \frac{\pi}{3},\ z \in \mathbb{C}$$

Thus, find z in polar form.

▼ Solution

Let $z = a + jb$

As $\left|\frac{z+j}{z}\right| = 4, \arg\left(\frac{z+j}{z}\right) = \frac{\pi}{3}$

$\frac{z+j}{z} = 4\angle\frac{\pi}{3}$ (notice the same

$\frac{z+j}{z} = 2 + j2\sqrt{3}$

$z + j = z(2 + j2\sqrt{3})$

$z - z(2 + j2\sqrt{3}) = -j$

$z(1 - 2 - j2\sqrt{3}) = -j$

$z(-1 - j2\sqrt{3}) = -j$

$z(1 + j2\sqrt{3}) = j$

$z = \frac{j}{1+j2\sqrt{3}}$

$z = \frac{1\angle\frac{\pi}{2}}{3.6056\angle 1.2898}$

$z = 0.28\angle 0.28$

Given that $z = a + jb$, and $w = \frac{z-1}{z+i}$,

Show that in general for z,

$$w = \frac{a^2 - a + b^2 + b + j(b - a + 1)}{a^2 + (b+1)^2}$$

Hence, find when w is purely real, purely imaginary, and |w| when $\arg(z) = \frac{\pi}{4}$

▼ Solution

$w = \frac{a+jb-1}{a+jb+i}$

$= \frac{(a-1+jb)(a-j(b+1))}{a^2+(b+1)^2}$

$= \frac{a(a-1)-j(a-1)(b+1)+jab+b(b+1)}{a^2+(b+1)^2}$

Use the binomial theorem to find $(\cos(\theta) + j\sin(\theta))^3$. Hence, show that

$$\cos(3\theta) = 4\cos^3(\theta) - 3\cos(\theta)$$

$$\sin(3\theta) = 3\sin(\theta) - 4\sin^3(\theta)$$

and

$$\frac{\sin(3\theta) - \sin(\theta)}{\cos(3\theta) + \cos(\theta)} = \tan(\theta)$$

and the exact value of $\tan(3\theta)$, given that $\sin(\theta) = \frac{1}{3}$.

▼ Solution

Given that $z_1 = -1 + j$ and $z_2 = \frac{1}{\sqrt{2}}\left[\cos\left(\frac{\pi}{3}\right) + j\sin\left(\frac{\pi}{3}\right)\right]$,

$$\text{Express } z_1 z_2 \text{ in the form of } a + jb$$

$$\text{Hence find } \tan\left(\frac{\pi}{12}\right) \text{ as } c + d\sqrt{3}, \text{ where } c, d \in \mathbb{Z}$$

$$\text{and smallest p such that}(z_1 z_2)^p \in \mathbb{R}^+, p \in \mathbb{Q}^+$$

▼ Solution

Given that $z_1 = -1 + \sqrt{3}aj$ and $z_2 = \frac{\sqrt{3}(a-1)}{3a+1} + j$, where $a \in \mathbb{R}, \neq -\frac{1}{3}$, show that

$$\tan(\theta) \text{ is constant, where } \theta = \arg\left(\frac{z_2}{z_1}\right).$$

and the two possible values of theta and the intervals of a for which they will occur.

▼ Solution

REVIEW:

Get to familiar; try to reach something that you know how to work with to get the answer.

See what looks close to familliar and try to reach that

Use direct definitions given to reach answer

If answer has out of place component (e.g. theta/2) try to reach that first

Use argand diagrams e.g. arg(-z)=arg(w) calculations

1<pi/4 = a+ja

https://www.revisionvillage.com/ib-math/analysis-and-approaches-hl/questionbank/number-and-algebra/complex-numbers/